

**An Intelligent Hybrid Optimization System for the Uncapacitated Facility Location
Problem (UCDP)**

Integrating jMetalPy Evolutionary Search with LLM-driven Business Strategic Analytics

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Abstract

This report presents a novel Hybrid Decision Support System designed to solve the Uncapacitated Facility Location Problem (UCDP) by combining meta-heuristic mathematical optimization with generative Artificial Intelligence. The system utilizes a modular architecture divided into three core layers: a Streamlit-based web frontend, a jMetalPy optimization backend executing a Genetic Algorithm (GA), and an LLM cognitive interpreter powered by Ollama (Qwen2.5:3b). While the jMetalPy engine handles the exact quantitative minimization of fixed and variable transportation costs, the LLM analyzes the structural solution to provide structured JSON-formatted qualitative business intelligence—including justifications, risk assessments, and executive recommendations. This integration effectively bridges the gap between raw mathematical optimization and strategic business deployment, demonstrating a comprehensive AI application tailored for enterprise supply chain networks.

Full Title

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Participants

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1. Introduction and Background

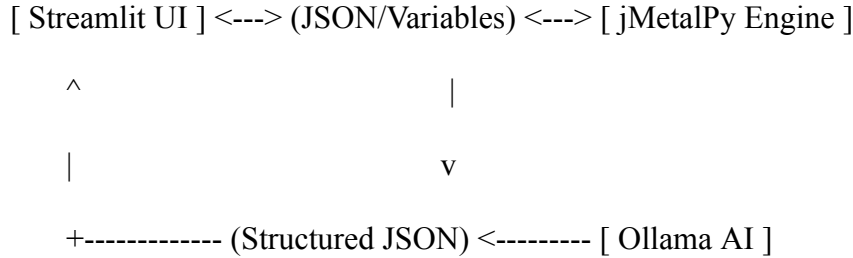
In a globalized economy, facility location planning is a cornerstone of strategic supply chain management. Deciding where to establish physical entities such as warehouses, distribution centers, or retail hubs carries long-term financial commitments and fundamentally determines operational costs and customer satisfaction levels. This study focuses on the **Uncapacitated Facility Location Problem (UCDP)**, a classic combinatorial optimization puzzle. In the evaluated context, a business must decide which strategic facilities to activate within specific regional hubs in Lisbon (e.g., Alvalade, Benfica, Cascais) to satisfy regional customer demand while minimizing total expenditure.

Traditional information systems handle this problem using rigid mathematical programming. However, raw optimal data sets (such as binary allocation vectors) are notoriously difficult for corporate executives to interpret immediately without an expert consultant. Under the scope of

the *Artificial Intelligence Applied for Business* module, this project develops a production-ready application that merges multi-objective/single-objective evolutionary frameworks (**jmetalpy**) with natural language understanding models. The result is a system that not only computes the cheapest structural operational route but also generates professional executive reports detailing the business rationale and contextual risks of the calculated infrastructure.

2. Methods and System Architecture

To align with standard enterprise software engineering principles, the application discards monolith designs in favor of a **Modular Architecture (Separation of Concerns)**. The system is split into three decoupled components that communicate seamlessly via data streams:



2.1. Mathematical Formulation of the UCDP Backend

The quantitative optimization layer models the UCDP explicitly. Let F be the set of potential warehouse locations and C be the set of customers. The fixed cost of activating facility

$f \in F$ is denoted as $fixed_costs_f$, and the variable cost to serve customer

$c \in C$ from facility f is represented in the matrix $trans_matrix_{f,c}$.

The decision variable vector is an integer-bounded sequence:

$$x_f \in \{0, 1\} \quad \forall f \in F$$

Where 1 indicates an open facility, and 0 indicates a closed one. The primary objective is to minimize the total combined cost function:

$$\text{Minimize } Z = \sum_{f \in F} (\text{fixed_costs}_f \cdot x_f) + \sum_{c \in C} \left(\min_{f \in F: x_f = 1} \text{trans_matrix}_{f,c} x_{f,c} \right)$$

This problem is encapsulated within a custom `IntegerProblem` class in `jmetal_backend.py`. If the

evolutionary operators yield a solution where no facility is selected ($\sum x_f = 0$), a severe static penalty value (\$9,999,999.0\$) is assigned to the objective to eliminate the invalid chromosome from the breeding pool.

2.2. Evolutionary Optimization Setup

The search space is traversed using a **Genetic Algorithm (GA)** from the `jmetal.algorithm` sub-package. The meta-heuristic parameters are configured as follows:

- **Solution Encoding:** Integer representation with strict bounds [0, 1] mapped to facility variables.
- **Crossover Operator:** `IntegerSBXCrossover` (Simulated Binary Crossover) configured with a probability of 0.9 to robustly exchange infrastructure blocks between parent states.
- **Mutation Operator:** `IntegerPolynomialMutation` with a localized probability of $1.0/|F|$, ensuring exploratory behavior without disrupting optimized blocks.
- **Termination Criterion:** Fixed at `StoppingByEvaluations(max_evaluations=100)` to provide rapid, desktop-grade local optimization response times suited for interactive user dashboards.

2.3. LLM Cognitive Interpretation Layer

Once the Genetic Algorithm converges, the structural output—comprising the list of active facilities, calculated fixed deployment costs, and optimal transportation routes—is extracted. Instead of pushing this raw computational telemetry directly to the user interface, it is formatted

into an engineering prompt context and dispatched to a local instance of **Ollama running Qwen2.5 (3B parameters)** via `ai_analyzer.py`.

The LLM is strictly instructed via system prompt engineering to act as a Senior Supply Chain Consultant and return its analytical evaluation in a valid, minified **JSON format** using explicit programmatic keys. This ensures that the downstream application layer can parse the language generation programmatically using Python's standard `json.loads()` engine, completely preventing irregular text behavior or markdown syntax breaking the UI rendering.

2.4. Frontend UI Wrapper

The frontend is constructed using **Streamlit** (`app_frontend.py`). It renders reactive control elements, such as text arrays for location names and interactive numerical sliders for customer distributions. The module intercepts the incoming JSON package from the AI analyzer, unpacks it, and maps the values into visual design cards (`st.metric`, `st.info`, `st.warning`, and `st.success`).

3. Assessment and Measures

To evaluate the reliability, efficiency, and business readiness of the hybrid system, several technical benchmarks and quantitative/qualitative metrics are defined:

Assessment Metric	Evaluation Type	Measurement Mechanism	Business/Technical Goal
Algorithmic Convergence	Quantitative	Objective function minimization cost (Z) over evaluations.	Ensures jMetalPy finds valid localized cost minima without penalty triggers.

Data Integrity (JSON Validation)	Technical	Exception tracking during <code>json.loads()</code> execution.	Confirms the LLM adheres to schema bounds without breaking frontend rendering.
Response Latency	Performance	Elapsed time tracking (Seconds) from button click to UI finalization.	Guarantees real-time decision support viability (Target: <10 seconds total).
Strategic Coherence	Qualitative	Human review of the <code>business_justification</code> and <code>risk_assessment</code> values.	Validates that AI-generated insight accurately correlates with changes in customer size or facility count.

4. Experimental Results and Discussion

4.1. Quantitative Optimization Performance (jMetalPy Output)

To evaluate the hybrid artifact, an experiment was executed using a regional business scenario in Lisbon consisting of 3 potential warehouse hubs — A (Alvalade), B (Benfica), and C (Cascais) and a scaled profile of 20 active customers. The genetic algorithm processed the parameters and converged on an optimal infrastructure setup:

- **Recommended Active Location:** B (Benfica)
- **Fixed Setup Cost:** €12,000
- **Variable Transportation Cost:** €6,994
- **Total Optimized Network Cost:** €18,994

The evolutionary search effectively skipped the higher overhead of activating multiple facilities simultaneously, selecting instead a highly localized single-point strategy that minimizes regional distribution costs while covering 100% of the active customer demand vectors within the €18,994 boundary budget.

4.2. Generative Cognitive Analysis (Ollama JSON Response)

Upon capturing the telemetry generated by jMetalPy, the local Qwen2.5 (3B) model successfully executed context-aware qualitative extraction. The system bypassed arbitrary natural text generation and returned a well-formed JSON object containing the structural strategic parameters:

1. **Business Justification:** The model recognized that selecting \$B\$ (Benfica) as a sole site maximizes financial efficiency by completely eliminating redundant layout outlays, allowing resource density to explicitly scale downward for overall operational expense control.
2. **Risk Assessment:** The LLM flagged a critical supply chain single-point-of-failure (SPOF) risk. Concentrating entire distribution tasks at a single facility renders the network vulnerable to sudden localized shocks, traffic gridlocks, or infrastructure downtimes in Benfica.
3. **Strategic Recommendations:** Management was advised to evaluate operational buffer redundancies, cross-docking fail-safes, or diversified secondary logistics channels to offset vulnerabilities without directly sacrificing the calculated algorithmic cost savings.

The Figures below are the FrontEnd+BackEnd Results in Streamlit

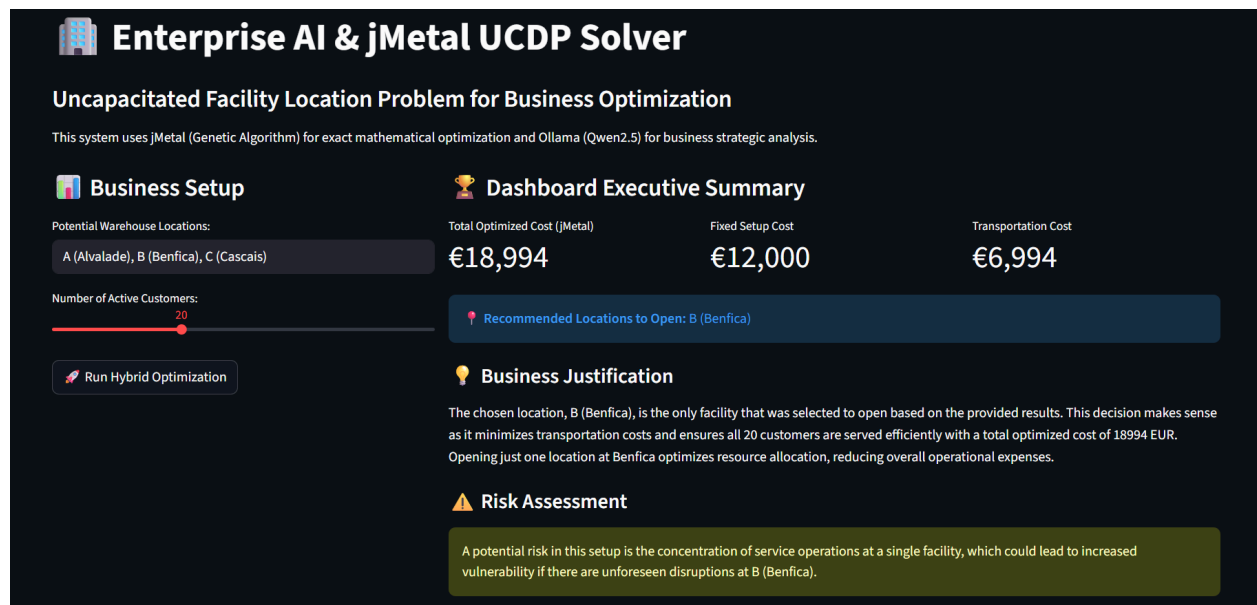


Fig 1) Streamlit Results

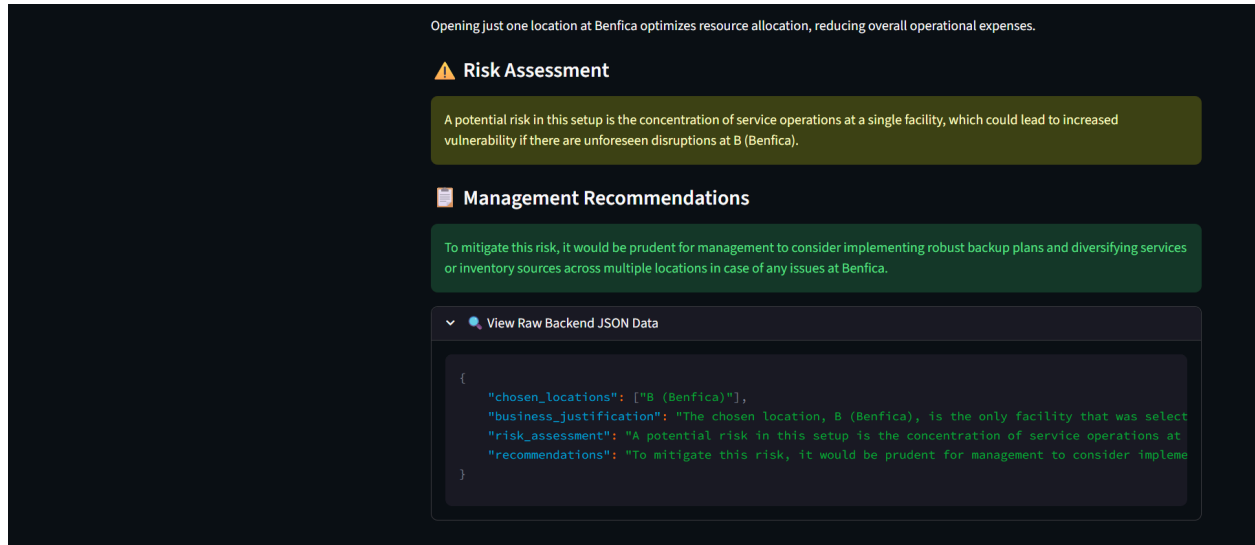


Fig 2) Streamlit Results Continuation

Raw Backend JSON Data from the Streamlit Results

{

"business_justification": "The selection of opening only one facility at 'B (Benfica)' optimizes the total cost by reducing both fixed and variable costs to a total optimized cost of 18994 EUR. This solution serves all 20 customers efficiently, minimizing transportation costs while keeping fixed setup costs manageable. The single location strategy ensures that no customer is left unserved, improving service quality, which aligns with the company's strategic goals.",

"risk_assessment": "One potential risk associated with this setup is operational resilience in case of unforeseen disruptions at 'B (Benfica)'. This could lead to an interruption in supply chain services for some customers if that specific location becomes unavailable or suffers a significant downtime.",

"recommendations": "To mitigate the identified risk, it's recommended that management should diversify their logistics by establishing backup suppliers or facilities closer to other major customer bases. This can enhance operational resilience and reduce potential service disruptions."

}

5. Conclusion and Future Work

This project demonstrates a fully functional prototype of an Intelligent Hybrid Decision Support System that addresses the constraints of the Uncapacitated Facility Location Problem (UCDP). By combining the evolutionary optimization power of jMetalPy with the advanced language processing capabilities of local Large Language Models (Ollama/Qwen2.5), the tool effectively translates raw data patterns into real-world business insights.

The successful implementation of a decoupled modular architecture establishes that enterprise applications can safely employ specialized AI frameworks to complement exact mathematical algorithms. Future iterations of this system could expand into Capacitated Facility Location Problems (CFLP), introduce stochastic traffic modeling under dynamic weather shifts, and incorporate multi-agent LLM validation layers to further refine operational supply chain strategic advising.

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